EHR Forecast

Final model report | 16 October 2026

A fixed LightGBM model predicts first-recorded diagnoses in the five years after a patient-specific anchor. The input is one patient's past synthetic EHR, represented by 512 selected features. No external data or pretrained weights are used.

Official split	Patients	Role
Train	2,791	Fit dictionaries, quality filters, selector and model
Validation	365	Score locked predictions; no final-model fitting
Test	358	Generate 40 probabilities; outcomes withheld

1. Representation and eligibility

Train and validation anchors are five calendar years before the final recorded encounter, using STOP or START if STOP is absent, as specified by the task. Test anchors are supplied. Future encounters serve only to define this required landmark. Predictors use events strictly before the anchor; the label is a condition's first-ever diagnosis in the following five-year window.

Source or field	Final treatment
Ten structured event tables	Past diagnoses, encounters, medication starts, procedures, labs, allergies, care plans, devices, immunizations and imaging examination facts are eligible.
Demographics and timing	Age and gender plus past counts, recency and observed elapsed time. Post-anchor stops are censored; death, administrative identifiers and future totals are excluded.
Annual QALY, DALY and QOLS	Excluded. A timestamp in the past does not establish that a yearly summary was computed using only past information.
Costs and reason fields	Excluded throughout the feature pipeline; later accounting or diagnosis context is not a verified historical predictor.
DNA, DICOM pixels and ECG payloads	Quarantined. DNA had phenotype-conditioned provenance concerns; local imaging/ECG payload-generation inputs were not fully certified. Structured examination facts remain eligible.

Numeric measurements retain units and missingness; conflicting simultaneous values are not silently averaged. Trajectories include levels, spread, change and time since measurement. Categorical histories, hashed text, visit relationships and clinical interactions extend the baseline. Duplicate projections preserve multiplicity where appropriate.

Training-only quality checks retain 16,181 of 36,327 candidate columns. Mutual information, regularized shallow-tree gain and sparse logistic coefficients produce a 512-column consensus set, with per-target protection. Three 80% training subsamples also produce a 256-column stability diagnostic. The deployed set remains the 512 consensus columns.

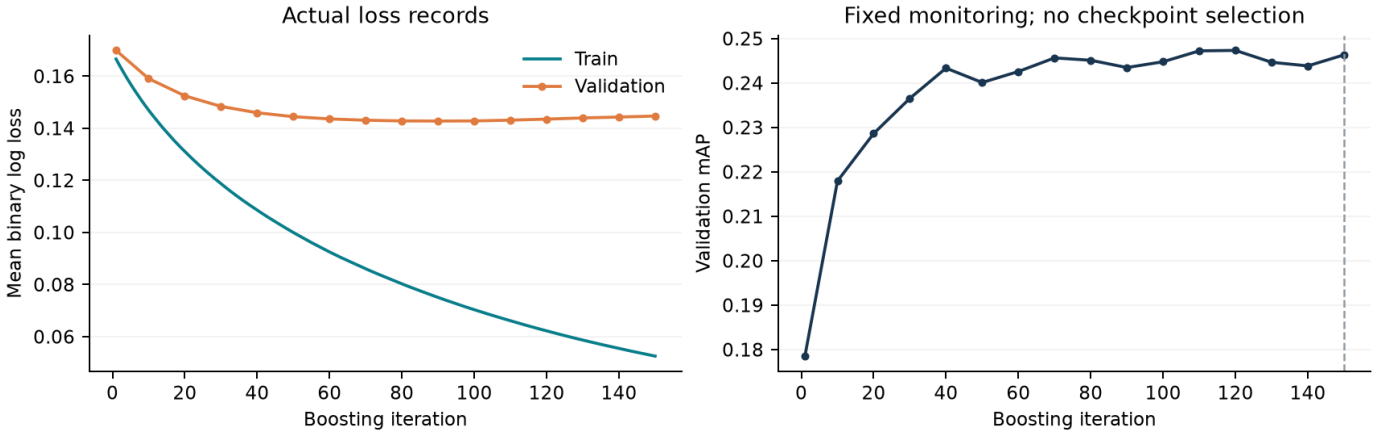
2. Model choice and training

The strict comparison used three training folds within the historical 1,675-patient fitting subset and a 558-patient development subset, all inside official train. Learned preprocessing and selection were refitted within each fold. Ten model families and two feature controls produced 48 actual fits; three blends were derived from out-of-fold predictions. Confirmation outcomes were not used to tune this comparison.

Model or control	OOF mAP	Dev mAP	Dev AUROC
LightGBM (512)	0.1728	0.2408	0.7590
LightGBM (earlier strict features)	0.1707	0.2349	0.7578
Nonnegative blend	0.1800	0.2331	0.7684
LightGBM (all quality columns)	0.1714	0.2330	0.7599
Equal top-three blend	0.1760	0.2273	0.7607
Equal ten-model blend	0.1732	0.2165	0.7694
CatBoost	0.1696	0.2112	0.7531
XGBoost	0.1649	0.2096	0.7505
TabM	0.1613	0.1976	0.7483
Elastic-net logistic	0.1620	0.1904	0.7458
ExtraTrees	0.1433	0.1814	0.7287
Random forest	0.1391	0.1810	0.7329
EBM	0.1380	0.1781	0.7359
MLP	0.1437	0.1754	0.7342
RBF SVM	0.1432	0.1748	0.7066

LightGBM was the strongest single model. The preselected blend did not improve development mAP. New versus earlier strict features improved the point estimate by 0.00590, but its paired 95% interval [-0.00607, 0.01925] included zero. These historical scores do not establish a stable improvement on new populations.

The final fit uses all 2,791 official train patients: 40 binary cross-entropy models, 150 trees, 31 leaves, learning rate 0.05, minimum leaf size 20, max_bin 127, L2=20, 0.8 row sampling every round, 0.8 column sampling, and seed 20261016 plus target index. There is no early stopping or ensemble. The recipe was fixed before final validation scoring.



Actual training loss and fixed-model validation replay are shown above. Validation points were computed after model and prediction locking; none was used to choose a checkpoint. The accepted model remains iteration 150. CSV values accompany this report.

3. Official-validation results

On 365 patients, mAP is **0.246334** and macro AUROC is **0.762251**. Prior-diagnosed targets are assigned probability zero before scoring. AP is averaged across all 40 targets; AUROC is averaged over targets with both classes (all 40 here). Hidden-test scores are unavailable.

Code	Condition	Pos.	AP	AUROC
444814009	Viral sinusitis	99	0.415	0.751
195662009	Acute viral pharyngitis	72	0.232	0.625
10509002	Acute bronchitis	61	0.296	0.667
88805009	Chronic congestive heart failure	29	0.339	0.824
230690007	Stroke	35	0.239	0.738
22298006	Myocardial Infarction	19	0.722	0.926
68496003	Polyp of colon	18	0.156	0.845
44465007	Sprain of ankle	16	0.046	0.507
53741008	Coronary Heart Disease	15	0.100	0.714
233604007	Pneumonia	17	0.623	0.983
43878008	Streptococcal sore throat	22	0.084	0.525
65966004	Fracture of forearm	16	0.111	0.664
126906006	Neoplasm of prostate	11	0.972	0.999
271737000	Anemia	15	0.126	0.794
64859006	Osteoporosis	12	0.088	0.811
443165006	Pathological fracture due to osteoporosis	14	0.305	0.890
49436004	Atrial Fibrillation	14	0.088	0.778
92691004	Carcinoma in situ of prostate	8	0.938	0.998
36971009	Sinusitis	9	0.043	0.561
75498004	Acute bacterial sinusitis	5	0.046	0.755
162573006	Suspected lung cancer	10	0.155	0.841
424132000	Non-small cell carcinoma of lung TNM stage 1	9	0.124	0.830
254637007	Non-small cell lung cancer	9	0.125	0.830
39848009	Whiplash injury to neck	8	0.028	0.513
26929004	Alzheimer's disease	13	0.116	0.849
263102004	Fracture subluxation of wrist	10	0.054	0.563
72892002	Normal pregnancy	10	0.604	0.991
713197008	Recurrent rectal polyp	10	0.223	0.899
62106007	Concussion with no loss of consciousness	7	0.091	0.725
15777000	Prediabetes	10	0.122	0.832
58150001	Fracture of clavicle	8	0.168	0.570
162864005	Body mass index 30+ - obesity	11	0.627	0.984
16114001	Fracture of ankle	8	0.176	0.700
70704007	Sprain of wrist	8	0.020	0.411
283371005	Laceration of forearm	7	0.040	0.526
314994000	Metastasis from malignant tumor of prostate	6	0.642	0.993
284551006	Laceration of foot	7	0.017	0.388
22325002	Abnormal gait	11	0.156	0.859
723857007	Silent micro-hemorrhage of brain	11	0.166	0.861
156073000	Fetus with unknown complication	5	0.230	0.971

The probabilities are risk rankings for this synthetic task, not validated clinical probabilities. Sparse positives make per-condition AP volatile; five to eleven positive cases support many of the rare-target estimates.

4. Interpretation and reproducibility

What the scores suggest

Prostate targets, pneumonia and obesity have strong ranking results, while several injuries and acute infections are much weaker. PSA-related history, examination context and measured physiological states are plausible predictive associations in this synthetic dataset. Feature importance does not establish causation or prove that the same signals would generalize to clinical records.

Neoplasm of prostate has AP 0.972 with only 11 positives; carcinoma in situ of prostate has AP 0.938 with eight. These strong point estimates still depend on small samples. Forearm laceration and wrist sprain have low AP and weak AUROC, consistent with limited historical signal and sparse events. No target-specific retuning was performed after seeing these results.

Generalization limits

Training loss falls to 0.05254, whereas final raw validation loss is 0.14470. The gap is consistent with fitting optimism; population differences and sample size can also affect the scores. The loss curve suggests diminishing validation gains late in training, but the prescribed 150-tree model was retained. These observations do not identify a causal percentage attributable to overfitting versus distribution change.

Official validation was examined in earlier project versions. It is therefore a reused evaluation set, not a virgin external confirmation. Earlier internal holdouts were merged into official train only after the recipe was fixed, so they cannot validate the final fit independently. The model uses one seed, and no claim of deployment robustness follows from this report. Payload exclusion lowers known provenance risk but does not certify that every source of leakage has been eliminated.

What can be reproduced

The repository provides raw-CSV preparation, fitted preprocessing, train-only feature selection, guarded RunPod training, native model export, prediction and post-lock validation scoring. Synthetic tests cover time boundaries, split isolation, prior masking, archive integrity, model replay and prediction locks. The model archive contains 40 native weights and fitted feature definitions, with no patient rows. Local predictions and caches stay outside Git.

Run the README commands with the supplied task dataset and pinned Python libraries. The release verification compares reconstructed training inputs, preprocessing, held-out probabilities and metrics with the accepted fit. Model selection and historical exploratory programs are archived locally; aggregate comparison results remain in the release. Exact final weights are provided, while clean-environment installation and a new cloud retraining are separate checks.

AI workflow

Codex assisted with schema inspection, leakage audits, feature and model implementation, test design, result checking and this report. The user set scope, approved cloud uploads and budgets, and chose the submission workflow. Acceptance relied on executed tests, hashes, native-weight replay and raw-input reconstruction rather than generated explanations. AI-assisted review does not replace independent evaluation.

Submission status and attribution

Code: github.com/LL333111/ehr-forecast-final. Hugging Face model publication and W&B; online synchronization remain pending. Actual offline W&B; runs 8ttc14tr (training) and kcb712gf (fixed validation replay) are retained locally; an online run URL must be added before submission. Test AUROC and mAP can only be supplied by the evaluator. Rename this file to your_name_results.pdf for the task.

Data: Coherent Synthetic Data Set, generated by Synthea. Copyright 2022 The MITRE Corporation, Public Release Case 21-1917; Creative Commons Attribution 4.0 International. The task's supplied ReadMe.md defines the split, landmark, label and submission rules. Raw data is not redistributed.